Teaching notes and solutions

> **Teacher note · Teacher notes.** In S2, immediately distinguish the index from the value of the term. In S4, have a loop executed by hand before Python. In S5 and S7, get students to say aloud "we assume at index n, we prove at index $n+1$". The sum and the divisibility come in S10, after Test 1: do not assess them in the first test. In S11, the search for the fixed point is an algebraic technique; the convergence theorem will be established in the next chapter. For weaker students, supply the property $P(n)$ and let them write the three steps; for the strongest, remove the hint about an auxiliary sequence.

Solutions to the activity

1. $u_0 = 100$ L. After one day, $0.8u_0$ L remain, then 30 L are added: $u_1 = 110$ L. Similarly, $u_2 = 0.8 \times 110 + 30 = 118$ L. For every $n \geq 0$, $u_{n+1} = 0.8u_n + 30$.

2. The equilibrium volume L would have to satisfy $L = 0.8L + 30$, that is $L = 150$ L. Let us prove $P(n)$: "$100 \leq u_n < 150$". At index 0, $u_0 = 100 \in [100, 150[$. Fix $n \geq 0$ and assume $P(n)$ is true. Since $0.8 > 0$,

$$110 = 0.8 \times 100 + 30 \leq 0.8u_n + 30 < 0.8 \times 150 + 30 = 150.$$

   Hence $100 \leq u_{n+1} < 150$ and $P(n+1)$ is true. By induction, the bounds hold for every $n \in \mathbb{N}$.

3. $v_{n+1} = u_{n+1} - 150 = 0.8u_n - 120 = 0.8(u_n - 150) = 0.8v_n$. The sequence (v_n) is geometric, with common ratio 0.8 and first term $v_0 = -50$. Thus $v_n = -50(0.8)^n$, then $u_n = 150 - 50(0.8)^n$.

4.
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def premier_jour():
    n, u = 0, 100
    while u <= 145:
        u = 0.8*u+30
        n += 1
    return n
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   The loop returns 11. Indeed, $u_{10} \simeq 144.63 < 145$ and $u_{11} \simeq 145.71 > 145$. Moreover, $u_{n+1} - u_n = 10(0.8)^n > 0$: the threshold was not crossed earlier. Calculating these two values guarantees the loop stops here, without yet requiring a convergence theorem.

5. The simulation only calculates a finite number of terms. The inductive step, together with the base case, establishes the bounds for all integers; it is this that rules out any excess over 150 L.

E1. $u_4 = 5 - 3 \times 4 = -7$. The sequence is arithmetic with common difference -3 and $u_{20} = -55$. The sum has 21 terms: $\sum_{k=0}^{20} u_k = \frac{21(u_0+u_{20})}{2} = \frac{21(5-55)}{2} = -525$.

E2. The sequence is geometric with common ratio 2 and first term $v_0 = 3$. Hence $v_n = 3 \times 2^n$. The sum has $n+1$ terms and equals $\frac{3(1-2^{n+1})}{1-2} = 3(2^{n+1} - 1)$.

E3. Let $P(n)$ be the statement "$3^n \geq 2n + 1$". At index 0, $3^0 = 1 = 2 \times 0 + 1$. Fix $n \geq 0$ and assume $P(n)$ is true. Multiplying by $3 > 0$, $3^{n+1} \geq 6n + 3$. Now $6n + 3 - (2n + 3) = 4n \geq 0$. Thus $3^{n+1} \geq 2(n+1) + 1$, which is $P(n+1)$. By induction, $P(n)$ is true for every $n \in \mathbb{N}$.

E4. For $n \geq 0$, $u_n = 1 - \frac{1}{n+2}$. We have $0 < \frac{1}{n+2} \leq \frac{1}{2}$, hence $\frac{1}{2} \leq u_n < 1$. Moreover,

$$u_{n+1} - u_n = \frac{1}{n+2} - \frac{1}{n+3} = \frac{1}{(n+2)(n+3)} > 0.$$

The sequence is strictly increasing, bounded below by $\frac{1}{2}$ and bounded above by 1.

E5. Let $P(n)$ be the statement "there exists an integer k such that $8^n - 1 = 7k$". For $n = 0$, $8^0 - 1 = 0 = 7 \times 0$: $P(0)$ is true. Fix $n \geq 0$ and assume $P(n)$ is true. There therefore exists an integer k such that $8^n - 1 = 7k$. Then

$$8^{n+1} - 1 = 8(8^n - 1) + 7 = 8 \times 7k + 7 = 7(8k + 1).$$

Since $8k + 1$ is an integer, $P(n+1)$ is true. By induction, 7 divides $8^n - 1$ for every $n \geq 0$.

Solutions to Homework 1

1. Let $P(n)$ be the statement "$2 \leq u_n < 6$". At index 0, $u_0 = 2$ satisfies the bounds. Fix $n \geq 0$ and assume $P(n)$ is true. Then $4 \leq \frac{u_n}{2} + 3 < 6$; in particular $2 \leq u_{n+1} < 6$. By induction, $P(n)$ is true at every index. We deduce $u_{n+1} - u_n = 3 - \frac{u_n}{2} = \frac{6-u_n}{2} > 0$. The sequence is strictly increasing.
2. The equation $L = \frac{L}{2} + 3$ gives $L = 6$. Let $v_n = u_n - 6$. Then $v_{n+1} = \frac{u_n}{2} - 3 = \frac{v_n}{2}$ and $v_0 = -4$. Hence $v_n = -\frac{4}{2^n}$ and $u_n = 6 - \frac{4}{2^n}$.
3. By linearity of sums and the geometric formula,

$$\begin{aligned} S_n &= 6(n+1) - 4\sum_{k=0}^{n} (\frac{1}{2})^k \\ &= 6(n+1) - 8(1 - 2^{-(n+1)}). \end{aligned}$$

4. $u_n > 5.99$ is equivalent to $\frac{4}{2^n} < 0.01$, that is $2^n > 400$. Since $2^8 = 256 \leq 400 < 512 = 2^9$ and (2^n) is increasing, the smallest index is 9.
5. The mean is less than 6 and its distance from 6 is

$$6 - \frac{S_n}{n+1} = \frac{8(1 - 2^{-(n+1)})}{n+1}, \qquad 0 < 6 - \frac{S_n}{n+1} < \frac{8}{n+1}.$$

   To make this distance less than a precision $\varepsilon > 0$, it suffices to take $n + 1 > \frac{8}{\varepsilon}$. This is a quantitative control of the distance, which does not yet require a theorem on limits.

Solutions to the textbook or past-paper work

Exercise 22. The four translations are, for every $n \in \mathbb{N}$: $u_{n+1} \geq u_n$; $v_n \geq 5$; $w_n \leq -1$; $t_n > 0$. Monotonicity compares two consecutive terms; a bound compares each term with one and the same number.

Exercise 25. We have $u_0 = -5$ and $u_{n+1} = \frac{2}{3}u_n - 1$.

Base case. $u_0 = -5 \leq -3$: the property is true at index 0.

Inductive step. Fix $n \in \mathbb{N}$ and assume $u_n \leq -3$. Multiplication by $\frac{2}{3} > 0$ preserves order, so $\frac{2}{3}u_n \leq -2$. Thus $u_{n+1} = \frac{2}{3}u_n - 1 \leq -3$.

Conclusion. By induction, $u_n \leq -3$ for every $n \in \mathbb{N}$. Next, $u_{n+1} - u_n = \frac{2}{3}u_n - 1 - u_n = -\frac{1}{3}u_n - 1$. The previous bound gives $-\frac{1}{3}u_n \geq 1$, hence $u_{n+1} - u_n \geq 0$. The sequence is increasing. The reversal of the inequality here comes from the negative factor $-\frac{1}{3}$.